

Ê PHASOR TEMPLATES (Air:  $\epsilon_r = \mu_r = 1$ )

Wavenumber per medium *nonmag.*  
 $k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c}$  (rad/m)  $\Rightarrow k_2 = k_1\sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$

Direction table  $u \in \{x, y, z\} = \text{axis perp. to boundary}$ 

| Inc dir    | Inc          | Refl         | Trans        |
|------------|--------------|--------------|--------------|
| $+\hat{u}$ | $e^{-jk_1u}$ | $e^{+jk_1u}$ | $e^{-jk_2u}$ |
| $-\hat{u}$ | $e^{+jk_1u}$ | $e^{-jk_1u}$ | $e^{+jk_2u}$ |

$+\hat{u}$  direction  $\Leftrightarrow$  Med 2 at  $u \geq 0$ . Reflected sign flips, transmitted matches incident.

General template (incident has pol.  $\hat{e}$ , amplitude  $a_0$ ):  
 $\hat{\mathbf{E}}^i = a_0 \hat{e} e^{\mp jk_1u}$  (V/m) (incident wave, Med 1)  
 $\hat{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm jk_1u}$  (V/m) (reflected wave, Med 1)  
 $\hat{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp jk_2u}$  (V/m) (transmitted wave, Med 2)  
(lossy med 2: replace  $e^{\mp jk_2u}$  with  $e^{\mp \alpha_2u} e^{\mp j\beta_2u}$ )  
Total in medium 1:  $\hat{\mathbf{E}}_1 = \hat{\mathbf{E}}^i + \hat{\mathbf{E}}^r$   
 $\hat{e}$  by polarization *plug into templates above*

| Pol.               | $\hat{e}$                            | Note              |
|--------------------|--------------------------------------|-------------------|
| Lin. $\hat{x}$     | $\hat{x}$                            | $E_y = 0$         |
| Lin. $\hat{y}$     | $\hat{y}$                            | $E_x = 0$         |
| Lin. at $45^\circ$ | $(\hat{x} + \hat{y})/\sqrt{2}$       | in phase          |
| RHC                | $\hat{x} - j\hat{y}$                 | $\delta = -\pi/2$ |
| LHC                | $\hat{x} + j\hat{y}$                 | $\delta = +\pi/2$ |
| General            | $a_x\hat{x} + a_ye^{j\delta}\hat{y}$ | elliptical        |

$\Gamma, \tau$  act on  $\hat{e}$  unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 —  $\Gamma, \tau, R, T$  (unitless) SUMMARY

Med. 1 ( $z < 0$ ) incident side; Med. 2 ( $z \geq 0$ ) transmitted.

|                    | Normal $\perp$ pol<br>( $\theta_i = 0$ )  | $\parallel$ pol                                                                                   |
|--------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------|
| $\Gamma$ refl. wav | $\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$ | $\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$ |
| $\tau$ trans. wav  | $\frac{2\eta_2}{\eta_2 + \eta_1}$         | $\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$                       |
| $\tau$ rel.        | $\tau = 1 + \Gamma$                       | $\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$                 |
| R refl. pwr        | $ \Gamma ^2$                              | $ \Gamma_{\perp} ^2$                                                                              |
| T trans. pwr       | $ \tau ^2 \frac{\eta_1}{\eta_2}$          | $ \tau_{\perp} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$                              |
| R+T total          | $T = 1 - R$                               | $T_{\perp} = 1 - R_{\perp}$                                                                       |

Notes:  $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$ ; nonmag.  $\Rightarrow \eta_2 / \eta_1 = n_1 / n_2$ .

Intrinsic impedance  $\eta_i$  *plug into table above*  
 $\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)} \quad \eta_0 = 120\pi \approx 377 \Omega$

Default  $\mu_r = 1$  (air, dielectric). Use  $\mu_r \neq 1$  only if problem says “ferromagnetic”/iron/ferrite/etc.

Low-loss correction  $\eta_c \propto \sigma / (\omega \epsilon) \ll 1$   
 $\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}} \left(1 + j \frac{\sigma}{2\omega \epsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$

For  $\Gamma/\tau$  just use  $\eta_0 / \sqrt{\epsilon_r}$ . The  $j$  term is for  $|\eta_c|, \theta_{\eta}$ . See LOSSY MEDIA — 5 CASES for all regimes.

POWER REFLECTION & TRANSMISSION

|                                                       |                                              |
|-------------------------------------------------------|----------------------------------------------|
| % reflected <i>always</i>                             | $\%P_r = 100  \Gamma ^2$                     |
| % transmitted <i><math>\eta</math>-ratio matters!</i> | $\%P_t = 100  \tau ^2 \frac{\eta_1}{\eta_2}$ |

Check:  $\%P_r + \%P_t = 100$ .

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*  
 $\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r,1}}{\epsilon_{r,2}}} = \sin \theta_1 \frac{n_1}{n_2}$

Multilayer chain *stack of slabs*  $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$   
 $\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$

Air–slabs–air shortcut *outer media identical*  
 $\theta_{\text{exit}} = \theta_{\text{incident}}$

*Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).*

LATERAL DISPLACEMENT

Beam offset through  $N$  slabs thickness  $t_i$ , refracted angle inside slab  $i$

Slab-indexed  $\theta_i =$  angle inside slab  $i$ ;  $t_i =$  slab thickness (m)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

The angle ( $\theta_i$ ) in the formula is always the one the ray makes inside that slab of height ( $t_i$ ).

LOSSY MEDIA — 5 CASES

Loss tangent  $\epsilon' = \epsilon_r \epsilon_0$ ;  $\epsilon_0 = 8.854 \times 10^{-12}$  F/m  
 $\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$

| Type                      | $\sigma / \omega \epsilon$ | $\eta_c, \alpha, \beta$                                                                                                                                                                                              |
|---------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lossless ( $\sigma = 0$ ) | 0                          | $\eta = \eta_0 / \sqrt{\epsilon_r}$ exact<br>$\alpha = 0, \beta = \omega \sqrt{\epsilon_r} / c$                                                                                                                      |
| Low-loss                  | $< 10^{-2}$                | $\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma}{2\omega \epsilon'}\right)$<br>$\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}, \beta \approx \frac{\omega \sqrt{\epsilon_r}}{c}$ |
| Quasi-cond.               | $10^{-2} - 10^2$           | exact: $(\eta / \sqrt{X}) e^{j\theta_{\eta}}$ see below                                                                                                                                                              |
| Good cond.                | $> 10^2$                   | $\eta_c \approx (1+j) \sqrt{\pi f \mu / \sigma}$<br>$\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$                                                                                                           |
| Magnetic                  | any                        | keep full $\sqrt{\mu_r / \epsilon_r} \eta_0$                                                                                                                                                                         |

For  $\Gamma/\tau$ : drop  $j$  correction; use  $\eta_0 / \sqrt{\epsilon_r}$ . Magnetic: keep  $\mu_r$ .

Low-loss quick params  $\sigma / (\omega \epsilon) \ll 1, \mu_r = 1$   
 $\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}$  (Np/m),  $\beta \approx \frac{\omega \sqrt{\epsilon_r}}{c}$  (rad/m),  $\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}}$  ( $\Omega$ )

Exact (quasi-cond)  $X = \sqrt{1 + (\epsilon'' / \epsilon')^2}$   
 $\alpha = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X-1}, \beta = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X+1}$   
 $\theta_{\eta} = \frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon'}$

RECTANGULAR WAVEGUIDE — MODES

TE mode  $E_z = 0, H_z \neq 0$   
TM mode  $H_z = 0, E_z \neq 0$   
Dimensions  $a \times b$  with  $a > b$  along  $\hat{x}, \hat{y}$ ; wave in  $\hat{z}$ .

$E_x$  form (TE) *read  $m, n$  from arguments*  
 $E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$

*coef. on  $x = m\pi/a$ ; coef. on  $y = n\pi/b$ .*

Identify TE $_{mn}$  vs TM $_{mn}$  *cos/sin pattern of  $E_x$  is identical for both!*

1. Read the problem statement (e.g. “a TE wave”  $\rightarrow$  TE).  
2. If  $m = 0$  or  $n = 0$  in the mode label  $\Rightarrow$  must be TE (TM forbids 0).  
3. Source field  $H_z(x, y)$  given  $\Rightarrow$  TE. Source field  $E_z(x, y)$  given  $\Rightarrow$  TM.

TE allows at least one of  $m, n \geq 1$  (other can be 0). TM needs both  $m, n \geq 1$ . That's why TE $_{10}$  exists but TM $_{10}$  doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

$e^{-j\beta z}$  factor omitted.  $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$ .  $\hat{E}$  in V/m,  $\hat{H}$  in A/m,  $Z$  in  $\Omega$ .

TE mode *generator:  $\tilde{H}_z$*   
 $\tilde{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$   
 $\tilde{E}_x = -\frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$   
 $\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$   
 $\tilde{E}_z = 0, \tilde{H}_x = -\tilde{E}_y / Z_{\text{TE}}, \tilde{H}_y = \tilde{E}_x / Z_{\text{TE}}$   
 $Z_{\text{TE}} = \eta / \sqrt{1 - (f_c/f)^2}$

TM mode *generator:  $\tilde{E}_z$*   
 $\tilde{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$   
 $\tilde{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$   
 $\tilde{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$   
 $\tilde{H}_z = 0, \tilde{H}_x = -\tilde{E}_y / Z_{\text{TM}}, \tilde{H}_y = \tilde{E}_x / Z_{\text{TM}}$   
 $Z_{\text{TM}} = \eta \sqrt{1 - (f_c/f)^2}$

TEM plane wave *for reference*  
 $\tilde{E}_x = E_0 e^{-j\beta z}, \tilde{E}_y = \tilde{E}_z = 0$   
 $\tilde{H}_y = \tilde{E}_x / \eta, \tilde{H}_x = \tilde{H}_z = 0$   
 $\eta = \sqrt{\mu/\epsilon} \text{ (}\Omega\text{)}, f_c \text{ N/A}, k = \omega \sqrt{\mu\epsilon}$

Properties common to TE/TM  
 $f_c = \frac{u_{p0}}{2} \sqrt{(m/a)^2 + (n/b)^2} \text{ (Hz)}$   
 $\beta = k \sqrt{1 - (f_c/f)^2} \text{ (rad/m)}$   
 $u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_c/f)^2}} \text{ (m/s)}$   
 $u_{p0} = 1/\sqrt{\mu\epsilon} \text{ (m/s)}$

CUTOFF FREQUENCY

$f_{mn}$  (a.k.a.  $f_c$  for that mode)  $m, n$  integers  $\geq 0$  (not both zero)

$f_c \equiv f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ (Hz)}$

Common cutoffs ( $a > b$ )

| Mode                       | $f_{mn}$                                         |
|----------------------------|--------------------------------------------------|
| $u_{p0}$                   | $c/\sqrt{\epsilon_r}$ (hollow: $u_{p0} = c$ )    |
| TE $_{10}$                 | $f_{10} = u_{p0}/(2a)$ (dominant)                |
| TE $_{01}$                 | $f_{01} = u_{p0}/(2b)$                           |
| TE $_{20}$                 | $f_{20} = u_{p0}/a = 2f_{10}$                    |
| TE $_{11}, \text{TM}_{11}$ | $f_{11} = \frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$ |

Propagation requires  $f > f_{mn}$ . In  $Z_{\text{TE}}/Z_{\text{TM}}$  and  $u_g, f_c = f_{mn}$  of the excited mode.

$\epsilon_r$  FROM DISPERSION

Given  $\omega, \beta$ , mode  $(m, n)$  *result is unitless*  
 $\epsilon_r = \frac{c^2}{\omega^2} \left[ \beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$  (unitless)

Inputs:  $\omega$  (rad/s),  $\beta$  (rad/m),  $a, b$  (m),  $c = 3 \times 10^8$  (m/s),  $m, n$  (unitless).

WAVE IMPEDANCE &  $H_y$

$Z_{\text{TE}} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} \text{ (}\Omega\text{)} \quad Z_{\text{TM}} = \eta \sqrt{1 - (f_c/f)^2} \text{ (}\Omega\text{)}$   
 $\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)} \quad H_y = \frac{E_x}{Z_{\text{TE}}} \text{ (A/m, TE mode)}$   
 $Z_{\text{TE}} > \eta$  always (denominator  $< 1$ );  $Z_{\text{TM}} < \eta$ .

GROUP VELOCITY & TRAVEL TIME

$$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \text{ (m/s)}$$

$$u_p = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}} \text{ (m/s)} \Rightarrow u_p u_g = u_{p0}^2 \text{ (m}^2\text{/s}^2\text{)}$$

$$t = L/u_g \text{ (s)} \quad \text{travel time through guide of length } L$$

Higher modes have lower  $u_g \Rightarrow$  they arrive later. Group delay spread = sum over excited modes.

BOUNDARY

$\eta_1, \eta_2$  — imp. ( $\Omega$ )  
 $\Gamma, \tau, R, T$  — coeffs (unitless)  
 $\tau = 1 + \Gamma$ ;  $R = |\Gamma|^2$   
 $k_i = \omega \sqrt{\epsilon_{r,i}}/c$  (rad/m)  
 $\alpha_2$  (Np/m),  $\beta_2$  (rad/m): lossy  
 $z = 0$  — boundary plane (m)  
 $\eta_0 = 120\pi \approx 377 \, \Omega$

SNELL / OBLIQUE

$\theta_i, \theta_t$  — angles from normal  
 $n_i = \sqrt{\epsilon_{r,i}}$  — index of refr.  
 $t_i$  — slab thickness (m)  
 $d$  — lateral disp. (m)  
plane of inc. = plane of  $\hat{k}$ +normal  
 $\perp$ :  $\mathbf{E} \perp$  this plane  
 $\parallel$ :  $\mathbf{E} \parallel$  this plane

WAVEGUIDE

$a, b$  — wide, narrow dim. (m);  
 $a > b$   
 $m, n$  — mode indices (int.)  
 $f_{mn}$  — cutoff (Hz)  
 $\beta$  — prop. const (rad/m)  
 $Z_{\text{TE/TM}}$  — wave imp. ( $\Omega$ )  
 $E_x$  — E-field comp. (V/m)  
 $H_y$  — H-field comp. (A/m)  
 $\eta = \eta_0/\sqrt{\epsilon_r}$

VELOCITIES

$u_{p0} = c/\sqrt{\epsilon_r}$  — bulk phase vel.  
 $u_p = u_{p0}/\sqrt{1 - (f_c/f)^2}$   
 $u_g = u_{p0} \sqrt{1 - (f_c/f)^2}$   
 $u_p u_g = u_{p0}^2$   
 $t = L/u_g$  — travel time  
 $c = 3 \times 10^8$  m/s

POWER & UNITS

$\%P_r, \%P_t$  — pwr refl./trans. (unitless)  
 $\%P_r = 100|\Gamma|^2$   
 $\%P_t = 100|\tau|^2 \eta_1/\eta_2$   
 $\%P_r + \%P_t = 100$   
 $\sigma$  (S/m),  $\epsilon$  (F/m),  $\mu$  (H/m)  
 $\epsilon_0 = 8.85 \times 10^{-12}$  F/m  
 $\mu_0 = 4\pi \times 10^{-7}$  H/m